Harsh Lembhe

+91-8625923006 lembheharsh0508@gmail.com LinkedIn GitHub

Education

Pune Institute of Computer Technology, Pune 2023 – 2027
Bachelor of Engineering in Artificial Intelligence and Data Science ***CGPA: 9.35/10***

Competitive Programming

- **Profiles**: Expert (1737) Codeforces | 4 Star (1931) CodeChef | Guardian (2202) LeetCode
- **Achieved top contest ranks**: 48 in CodeChef Starters 241, 109 in Biweekly Contest 186, LeetCode, & 175 in Codeforces Round 1107 (Div. 3).
- **Secured AIR 1458** in IICPC's CodeFest'25 Prelims, organized by top IIT's.
- **Solved 3000+** programming problems across all platforms, sustaining a 500+ day continuous practice streak on multiple coding platforms.

Projects

FinRAG: RAG Framework for Financial Documents | *React, FastAPI, LangChain, Pinecone, Docker* | **Live** | **GitHub**

- **Architected** an end-to-end RAG service for financial PDFs (10-K filings, annual reports), returning cited, page-grounded answers to eliminate manual document search.
- **Built** a two-stage hybrid retrieval pipeline - dense (**Pinecone**) + sparse (**BM25**) search followed by cross-encoder re-ranking to improve retrieval precision on numerical, multi-hop financial queries.
- **Engineered** an async ingestion pipeline: API enqueues tasks to **Celery** workers running in **Docker** containers, using **Redis** as broker and as a separate cache for repeated LLM query responses.
- **Integrated** live market data (**yfinance**), a real-time stock news feed, and a self-authored market glossary for retail traders.

Forge IDE: CP for PICT Coders League | *Next.js, Node.js, PostgreSQL, Redis, Piston API* | **Live** | **GitHub**

- **Architected** a full-stack browser **IDE** for competitive programming contests using **Next.js** and **Node.js**, routing submissions to the **Piston API** for sandboxed multi-language execution.
- **Designed** a **"Reverse Coding"** mode with a custom execution wrapper around Piston that runs contestant inputs through a hidden reference solution for logic inference.
- **Built** an automated ingestion pipeline that web-scrapes problems from external platforms, and cached live leaderboard rankings in **Redis** to avoid recomputing standings from **PostgreSQL** on every request.
- **Engineered** a real-time collaborative whiteboard and chat system using **Socket.io**, broadcasting draw/cursor events room-wise to synchronize state across all connected clients, alongside **LiveKit**-powered video/voice for **200+ concurrent club members**.

Leadership

PICT ACM
Event CP Lead

- **Mentorship**: **Led** a SIG (Special Interest Group) on competitive programming, upskilling 200+ students.
- **Event Management**: **Spearheaded** the full **Compute and Compete** event, curating problems for Reverse Coding and BugOff and seamlessly managing execution end-to-end for 1000+ students across multiple colleges all over India.

PICT INC
Problem Setter Lead

- **Problem Setting**: **Curated** 10+ problems of varying difficulty levels and led the problem-setting team for an international-level competitive programming contest.
- **Scale**: **Drove** participation of 1000+ contestants from across India, including Masters and Candidate Masters.

Technical Skills

Languages: Python, Java, C++, SQL, JavaScript (ES6+)
Technologies: Express.js, React.js, Node.js (MERN), FastAPI, Flask, Docker, Redis
Database: MongoDB, MySQL, PostgreSQL, Prisma, Drizzle ORM, Pinecone, FAISS (Vector DB)
Soft Skills: Communication, Leadership, Team Collaboration

Technical Achievements

- **Won DataQuest** Model Building Competition by PICT ACM, outperforming **100+ participants**.
- **Finalist, Top 100** at **Suraksha Cyber Hackathon** by **Canara Bank** (International Level).
- **Secured 3rd place** in AIT's **CODEFT 5.0**, outperforming 600+ competing coders.
- **Finalist** at **CodeUncode** by DJSCE Mumbai.